🌱 **Smart Garden Simulation** — *Quick Reference Guide*

SMART GARDEN USER MANUAL

A compact guide to every feature in the Smart Garden simulation UI

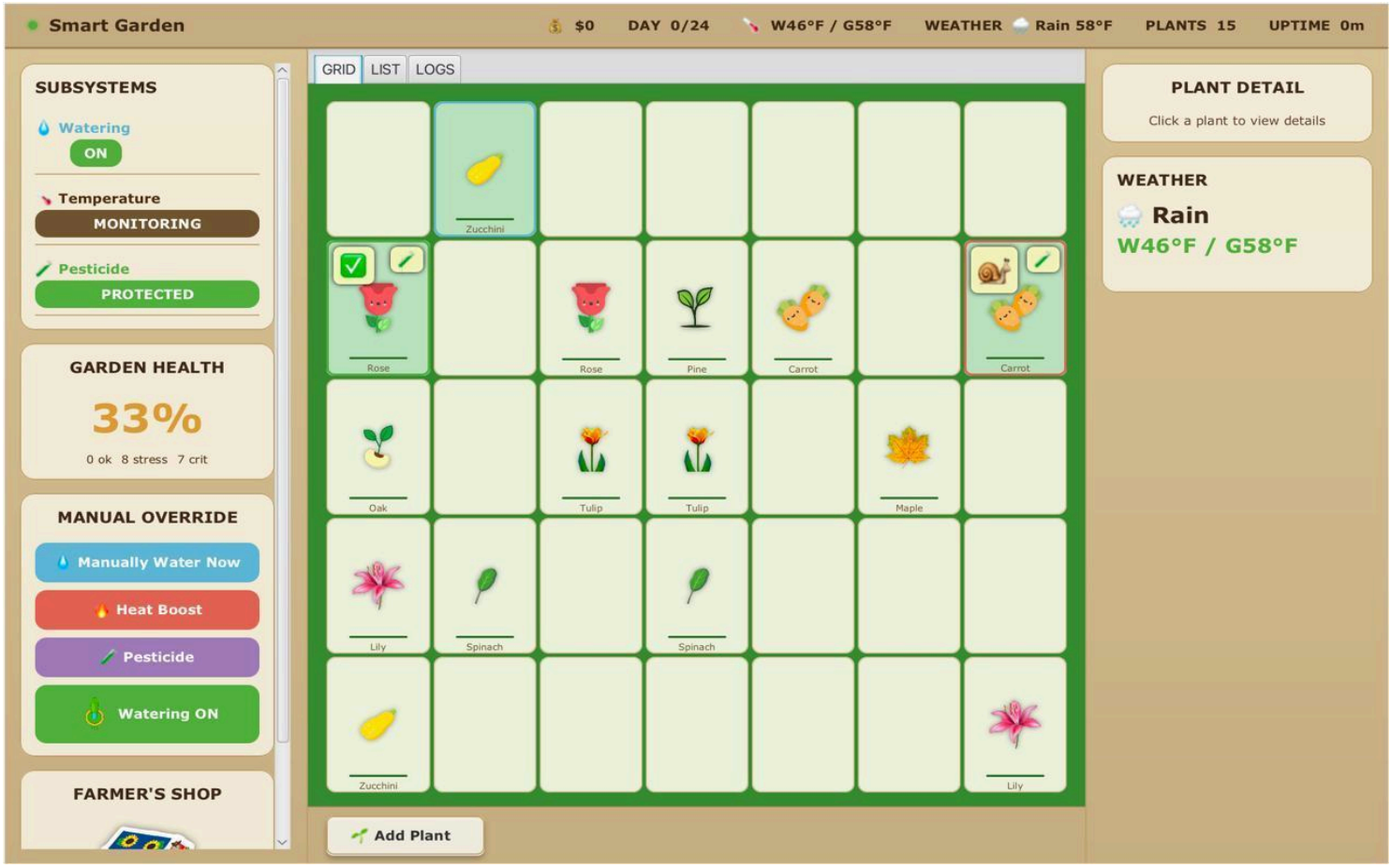

Figure 1: The Smart Garden interface — all major areas labeled below

📊 Header Bar

The top bar shows real-time garden status at a glance:

Item	Example	What It Means
🟢 Status	Green dot + “Smart Garden”	System is running normally
💰 Money	$0	Your current balance from selling harvested crops
📅 Day	DAY 8	Current day (Total game days the simulation is running)
🌡️ Temp	W46°F / G58°F	W = Weather (outside) temp, G = Greenhouse (inside) temp
⛅ Weather	Rain 58°F	Current weather condition and temperature
🌿 Plants	PLANTS 15	Total number of living plants in your garden
⏱ Uptime	UPTIME 0m	How long the simulation has been running

⚙️ Left Sidebar — Subsystems & Controls

SUBSYSTEMS

These show the current status of each automated system:

Subsystem	Status Shown	What It Does
💧 Watering	ON / OFF	Monitors soil moisture and auto-waters plants when their water level drops below their threshold. When ON, sprinklers activate automatically for thirsty plants.
🌡️ Temperature	MONITORING	Tracks weather and greenhouse temps. Automatically heats or cools plants when temperature goes outside their preferred range.
🧪 Pesticide	PROTECTED	Detects pests on plants and auto-sprays pesticide. Shows PROTECTED when pest treatment is active, READY when idle.
🏠Farmer's Shop		Located at the bottom of the left sidebar. Plants that reach full health are auto-harvested and appear here as inventory. Once you accumulate 10 items, they auto-sell and the money is added to your $balance in the header. Floating "+$X" animations appear on harvested cells.

GARDEN HEALTH

Display	Meaning
33%	Overall percentage of garden health. Calculated from all living plants' individual health values.
0 ok 8 stress 7 crit	Breakdown: OK = healthy plants, Stress = plants under moderate pressure, Crit = critically damaged plants near death.

MANUAL OVERRIDE

These buttons let you intervene manually. The system will warn you if your action conflicts with the automated plan.

Button	What It Does When Clicked
💧 **Manually Water Now**	Immediately triggers a watering cycle for ALL plants, regardless of whether they need it. Use this if you notice plants drying out faster than the auto-system can respond. Warning: over-watering can waste resources.
🔥 **Heat Boost**	Forces the heating system to raise the greenhouse temperature immediately. Useful during sudden cold snaps when plants are taking cold damage. The system will show a warning if the greenhouse is already warm enough.
🧪 **Pesticide**	Manually triggers a pesticide spray across affected plants. Use if you see pest icons on grid cells and want to speed up treatment instead of waiting for the auto-cycle. Each use consumes pesticide from the supply.
💦 **Watering ON/OFF**	Toggles the automatic watering subsystem on or off. When OFF, no auto-watering occurs — you must use "Manually Water Now" to water. When ON, the system resumes its scheduled checks. Useful for rainy days when you want to conserve water.

🌱 Garden Grid (Center)

The main 7×7 grid is the heart of the simulation. Each cell can hold one plant.

Element	Description
Plant image + name	Each occupied cell shows the plant's current growth-stage image and its name below. Images change as plants grow through lifecycle stages.
Health bar (below name)	A small horizontal bar under each plant name. Shows green when healthy, yellow when stressed, red when critical.
Pest icons (corner)	When a pest attacks, its icon (snail, crow, etc.) appears in the cell corner. Example: the snail icon on the Carrot cell in the screenshot shows an active pest.
Status checkmarks	Green checkmark = plant recently watered/treated. Pencil icon = pesticide recently applied to this cell.
Red border on cell	Indicates the plant is under active pest attack or in critical health condition.
Light tint on cell	Light blue/green tint = cell was recently watered. Light red = cell is under stress.
Empty cells	Dark green cells with no content are available for planting.

🗂️VIEW TABS (top of grid):

Element	Description
GRID	visual garden view
LIST	text list of all plants
LOGS	Click the LOGS tab at the top of the grid to view the full event log. ➢ Every action is timestamped with the day number: weather changes, watering events, pest attacks, pesticide sprays, plant growth, harvests, and deaths. ➢ The format is "Day: [N] [event description]". Logs are the primary tool for verifying that all subsystems are functioning correctly.

🌱 Add Plant Button

Element	Description
🌱 **Add Plant Button**	Click the "Add Plant" button below the grid to open a plant selection menu. Choose from Trees (Oak, Maple, Pine), Flowers (Rose, Tulip, Lily, Sunflower, Jasmine), or Vegetables (Carrot, Tomato, Lettuce, Zucchini, Spinach). The new plant is placed in a random empty cell with a grow-in animation

ℹ️ Right Sidebar

Element	Description
PLANT DETAIL	Click any plant on the grid to see its detailed info here — species, health, water level, temperature preference, and current status.
WEATHER PANEL	Shows the current weather condition (Rain/Sunny), plus W (weather/outside) and G (greenhouse/inside) temperatures. During rain, plants get natural water so the sprinkler system may skip cycles.

🐛 Pest Quick Reference

Pest	Rounds	Damage	Counter	How You'll See It
Aphids	2	Low–Med	Auto pesticide	Green bug icon on cell + red damage number
Slugs	2	Moderate	Auto pesticide	Snail icon on cell (visible in screenshot!)
Rat	3	High	Auto pesticide (slow)	Rat icon on cell + needs 3 treatment rounds
Locust	3	High	Auto pesticide (slow)	Locust swarm icon on cell
Crow	1	Moderate	Scarecrow (auto)	Crow icon + scarecrow pop-in animation

⏰ Day Cycle & Timing

Timing	Detail
1 garden day = ~ 1 Hour real time	The simulation progresses through days automatically. Each new day triggers weather, temperature, pest, and watering events.
Full run	The DAY counter in the header tracks progress. At the end of the run, review the LOGS tab for complete event history.

💡 Tips for Best Results

- The garden survives on its own! You only need to intervene via Manual Override if things go wrong.
- Watch the Garden Health % - if it drops below 20%, consider manual watering and heat boost.
- Rainy days = free water.
- Check the LOGS tab regularly - every event should be visible.
- Rats and Locusts take 3 rounds to kill - don't panic if pests persist for a few days.